

Queue Class - 3

Special class

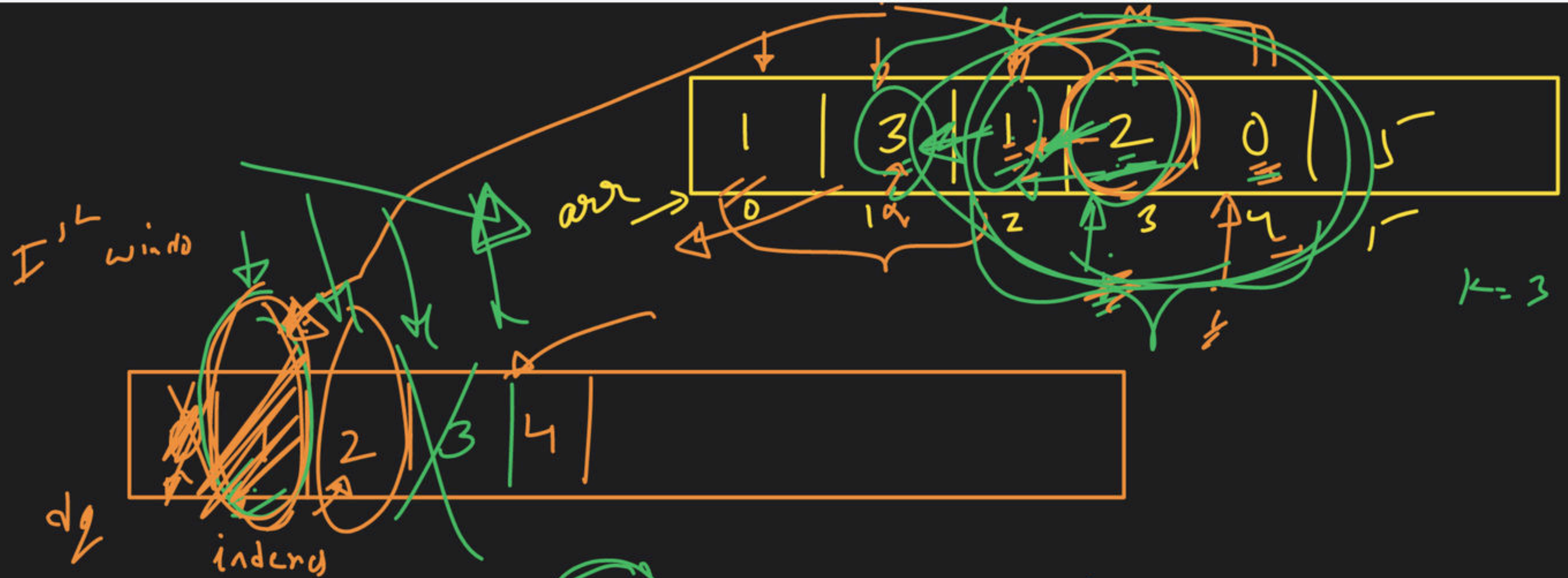
→ Sliding Window Maximum

→ T_0^+ →
 T_1^- array
then
→ sum
→ add
→ ans

[3 7 5]
↑ ↑
~~9~~ ~~4~~

[0 -1 6]
↑ ^ ↑
 α α

~~9~~ $\geq$ and α
< arr[i]
pop

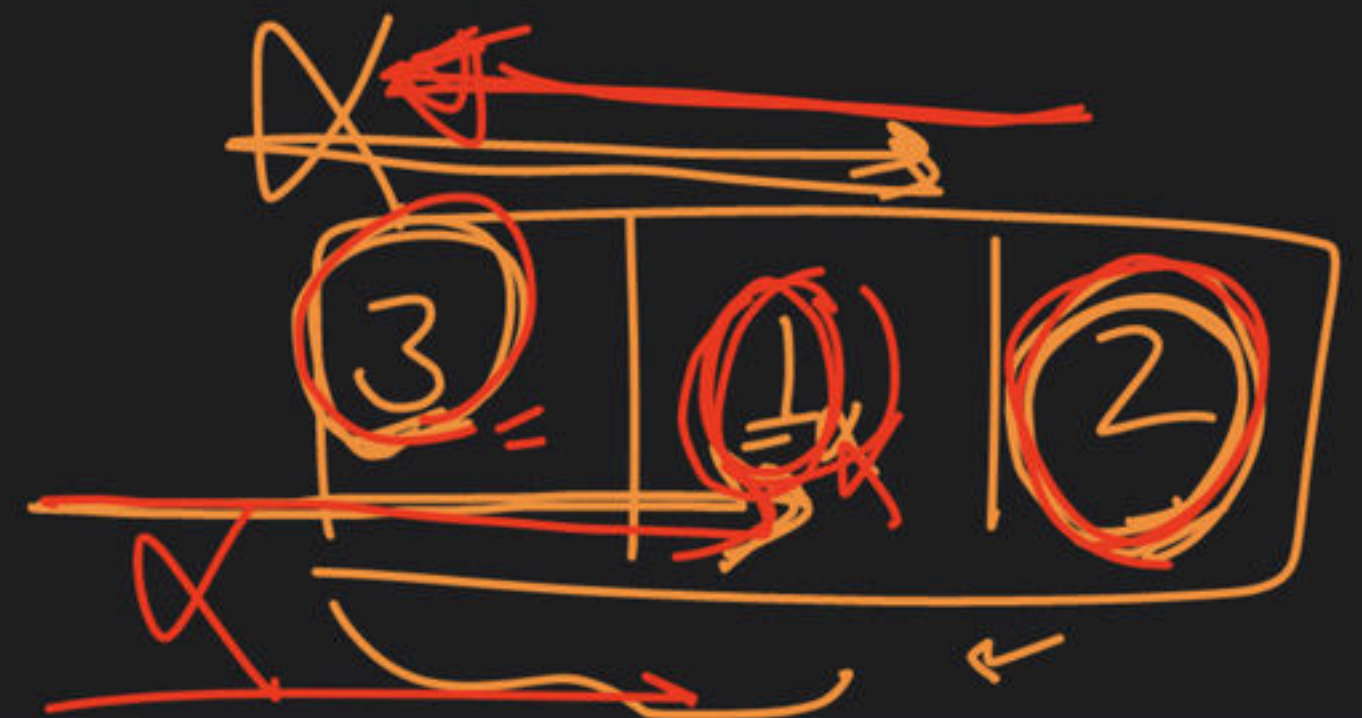
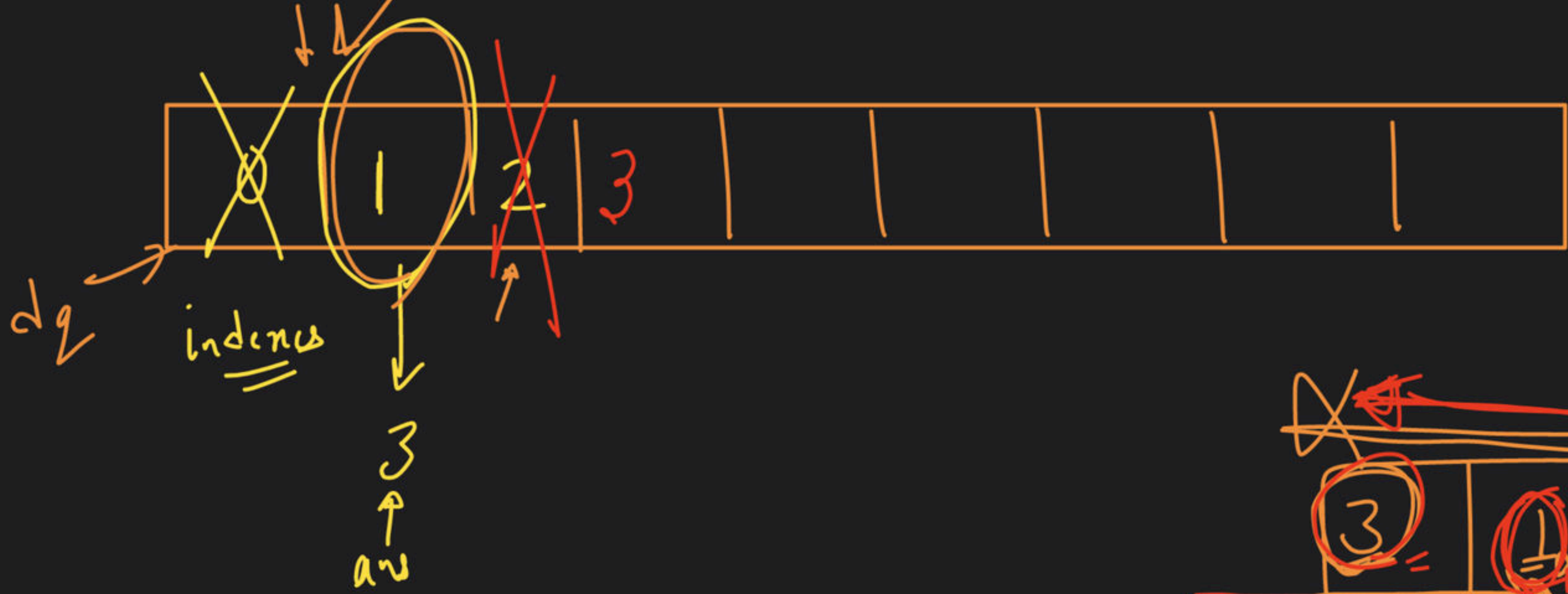
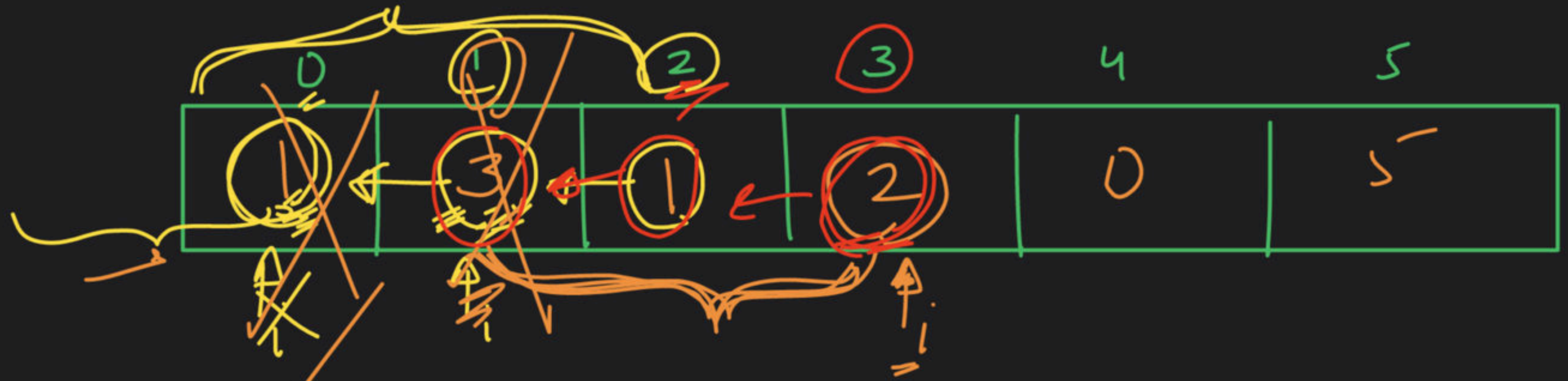


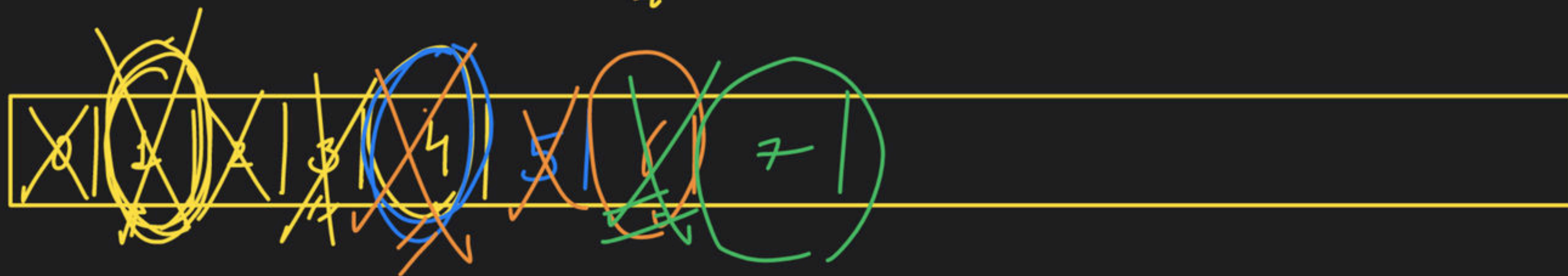
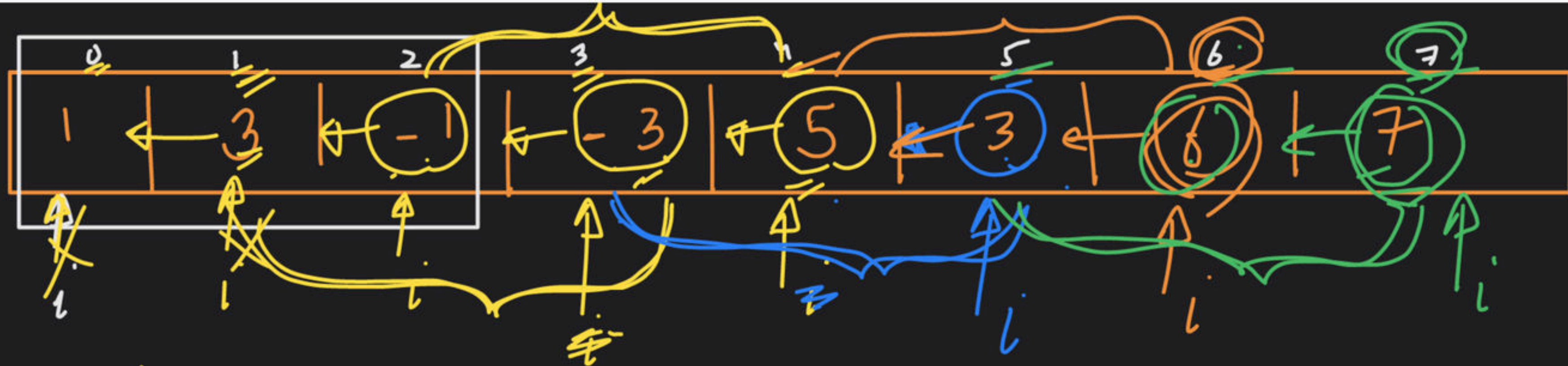
T^L window $\rightarrow$ `[3]` `[3]` `[1]`

$1 < 4-3H$
 $1 < 2 \rightarrow T$

Run $\rightarrow$ Π $\rightarrow$ `sum`
 $\rightarrow$ `addite`
 $\rightarrow$ `ans`

Π $\rightarrow$ `run`
 $\rightarrow$ `add`
 $\rightarrow$ `ans ->`





dg

$$\text{I} \rightarrow 3$$

$$\text{II} \rightarrow 3$$

$$\text{III} \rightarrow 5$$

$$\text{IV} \rightarrow 5$$

$$\text{V} \rightarrow 6$$

$$\text{VI} \rightarrow 7$$

$$\{3, 3, 5, 5, 6, 7\}$$



freq

char	int
a	2
c	1
d	2
e	1

a a c d d e

↑ ↑ ↑ ↑ ↑

i i i

~~a~~ | ~~a~~ | (c) | d | d | e

↑ ↑ ↑

ans → a#c c c c

① $\downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow \downarrow$
aa b e d a b b a j a b b a

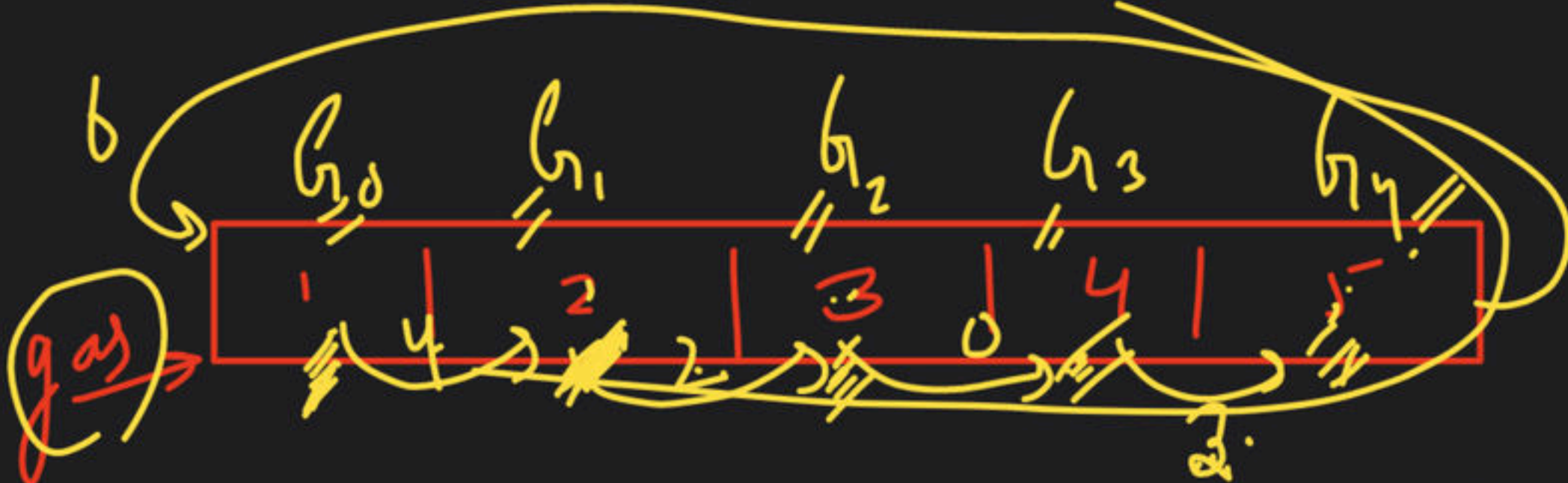
②

a	a	b	a	d	a	b	b	a	j	a	b	b	a
---	---	---	---	---	---	---	---	---	---	---	---	---	---

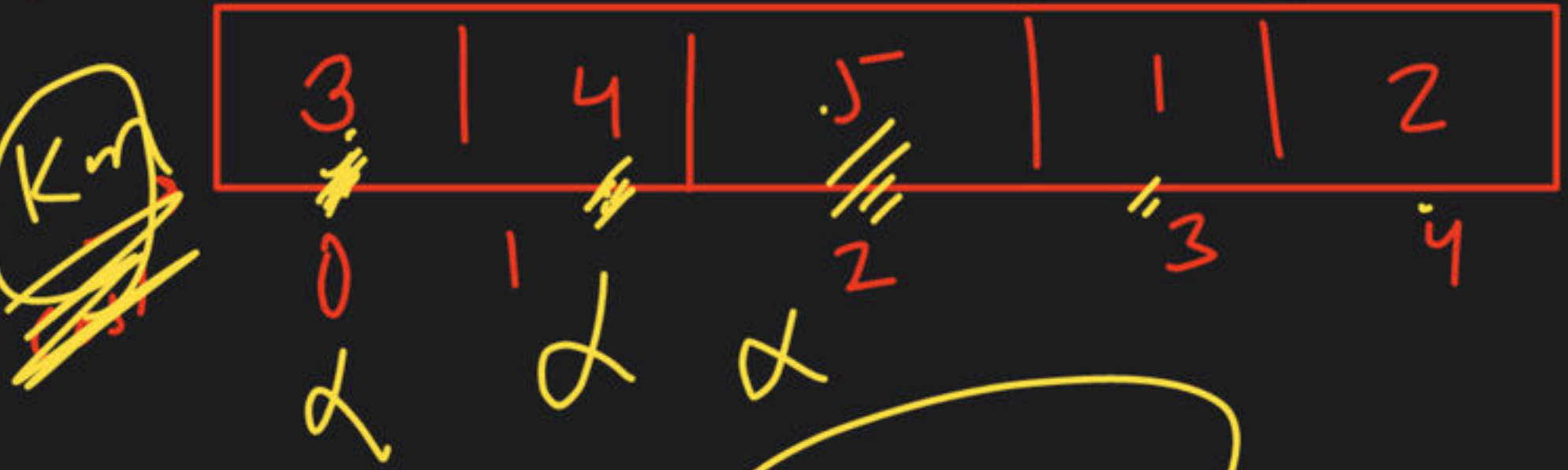
↑ ③ ↑ ↑ ↑ ↑

ans $\rightarrow$ a#b b b b d d d d d d d d

char	int
③ a	2 3 4 5 6 7
b	1 2 3 4 5
d	①
j	1



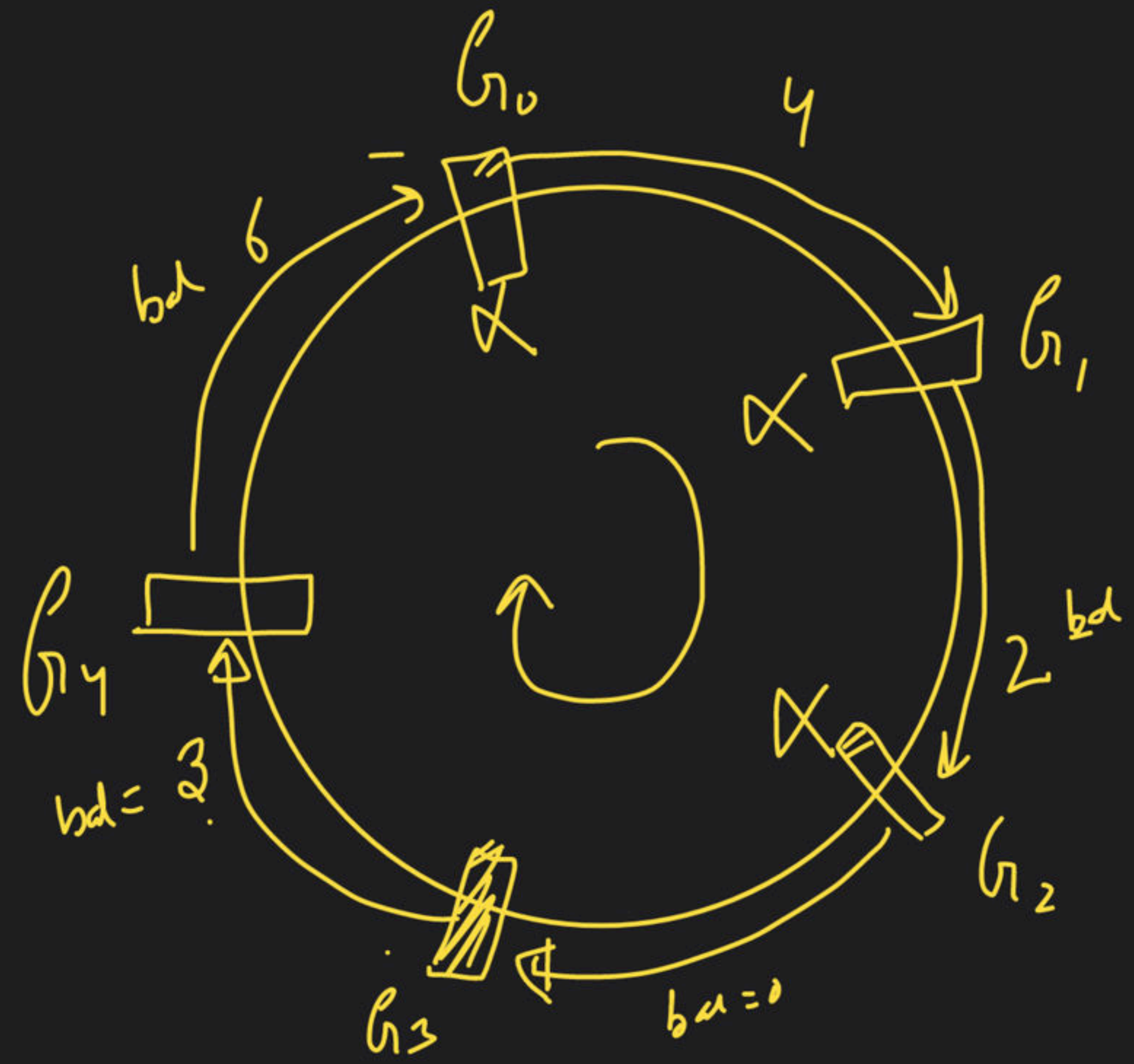
1 liter gas $\rightarrow$ 1 km

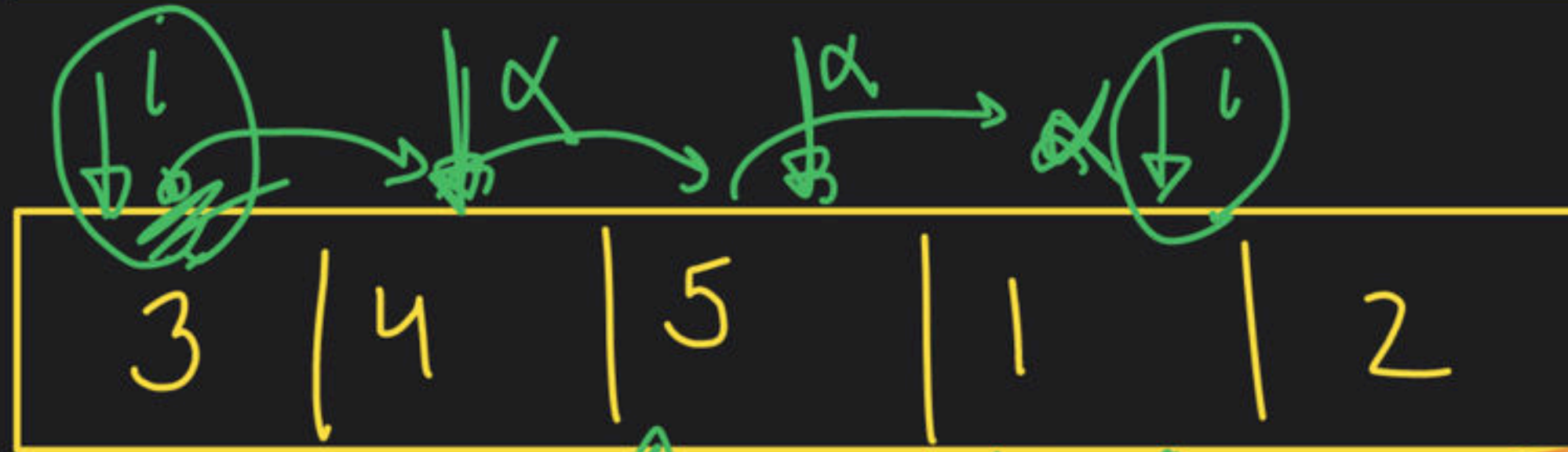


ans = 3 index

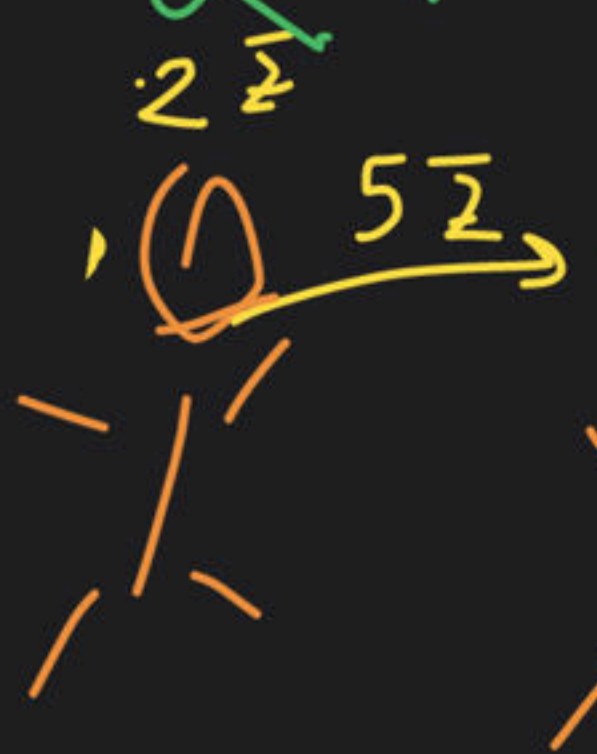
Brute force

$O(n^2)$





$8-i$



$3 \pm$

$8 \pm$



10 $\pm$ need

Why?

i

gas →

km →



2
deficit

2
deficit

2
deficit

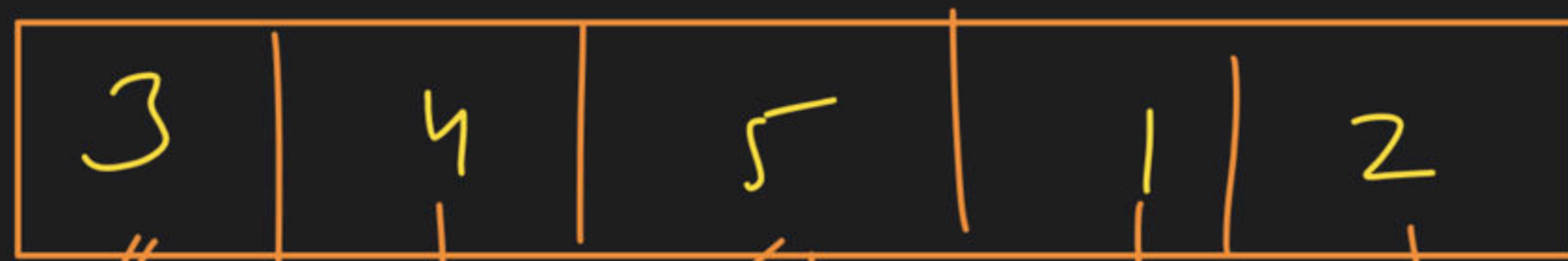
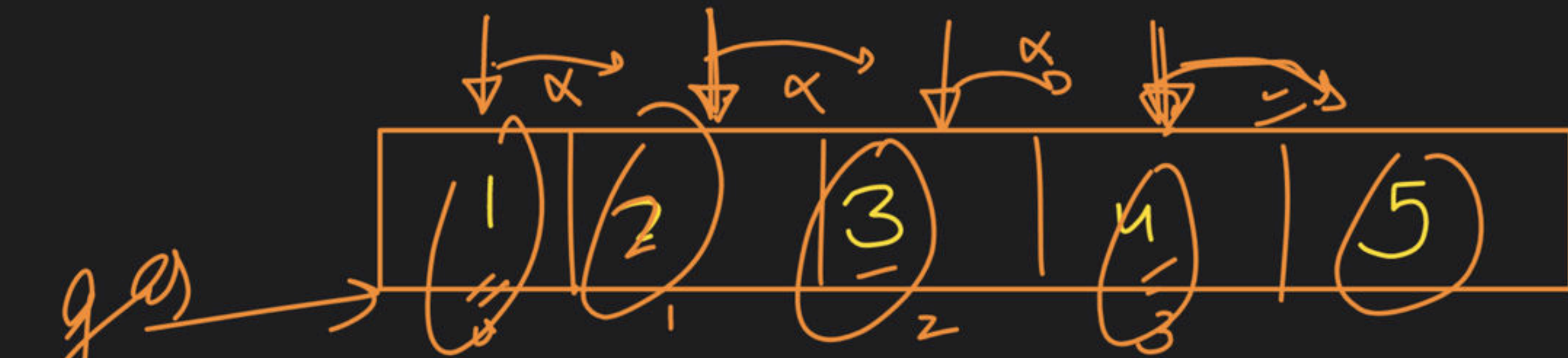
bal = 3

bal = 1

+ 1
bal

Total deficit = 6 lit





Km / out

$gas < Km$

$deficit = 2$

$gas < Km$

$deficit = 2$

$gas < Km$

$deficit = |3 - 1| = 2$

$gas > Km$

$bal = |4 - 1| = 3$

$gas > Km$

$bal = |5 - 2| = 3$

~~deficit = 6~~
~~balance = 6~~
~~start = 3~~

deficit = 6, balance = 6, start = 3

if (bal - deficit $\geq$ 0)

↳ circle complete $\rightarrow$ return start

else

↳ circle can't be completed $\rightarrow$ return -1





































